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Operations

AIRFIELD DAMAGE REPAIR (ADR) CAPABILITIES

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This publication supports Air Force Instruction (AFI) 10-209, *RED HORSE Program*, AFI 10-210, *Prime Base Engineer Force Program*, and Air Force Doctrine Publication 3-34, *Engineer Operations*. It provides engineering, operations, and logistics planners an introduction and overview of contingency Airfield Damage Repair (ADR) capabilities. This publication applies to all DAF civilian employees and uniformed members of the Regular Air Force, the Air Force Reserve, and the Air National Guard civil engineer units. This publication does not apply to the United States Space Force. Refer recommended changes and questions about this publication to the Office of Primary Responsibility using the Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command and Major Command publications/ forms managers. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program* and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the DAF.

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. In addition to updating terminology and references, it has been reorganized as a planning document to provide an overview of ADR capabilities and how each supports the Air Force Force Generation (AFFORGEN) Force Elements (FE) of Open the Airbase, Command and Control, Establish the Airbase, Operate the Airbase (including sustain and recover), Robust the Airbase, and Demand Force Teams. Hands-on tactics, techniques and procedures (TTPs) were transferred to Air Force TTPs (AFTTPs) and referenced within this document where appropriate.

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Chapter 1

INTRODUCTION

1.1. Overview. The objective of Airfield Damage Repair (ADR) is to enable combat forces to operate across the full range of military operations by achieving the desired primary effect of rapid projection and application of Joint US combat airpower worldwide, from both enduring and non-enduring bases, in highly contested and non-contested environments. Airfield damage repair, as related to Air Force civil engineers, encompasses capabilities required to rapidly repair aircraft operating surfaces in order to open, establish, operate (including sustain and recover), and provide robust airfield operations, whether the airfield is seized from the enemy, owned by the United States, or offered for use by a host nation. This chapter overviews ADR and ADR Unit Type Codes (UTC), **Chapter 2** provides a brief description of each UTC's capability, and chapters 3 through 6 describe capabilities performed within the Air Force Force Generation (AFFORGEN) Force Elements (FE). Additional resources for engineer doctrine and standards may be found using the engineer Reachback links in **Attachment 2**.

1.1.1. Engineers restore deteriorated airfields, sustain temporary repairs while operating at high operational tempo on substandard and/or aged pavements, and rapidly recover damaged airfields after an enemy conventional attack. The pavement repairs support a mix of rotary-wing, fighter, tanker, and airlift missions from the same operating surfaces. To reduce or eliminate impact to high operations tempos, repairs must be durable and sustainable.

1.1.2. This pamphlet provides engineering, operations, and logistics planners an overview of USAF civil engineer capabilities required to accomplish the ADR mission. Specifically, it describes ADR capabilities required for opening, establishing, operating, sustaining, and recovering airfields. It references source documents that provide tactics, techniques, and procedures for craftsmen to execute ADR capabilities described in this pamphlet. Hands-on, step-by-step repair procedures are covered in their respective technical orders (TO), or Air Force Tactics, Techniques, and Procedures (AFTTPs) listed at the end of this chapter or referenced in this pamphlet. **Table 1.1** lists ADR-related UTCs and selected Force Elements and sustainment activities the UTCs fall within.

Table 1.1. ADR Related UTC Roles across Selected AFFORGEN Force Elements.

UTC	Title	Open	Establish	Operate (Sustain & Recover)	Robust
4F9AD	Airfield Pavements Evaluation Team	X		X	
4F9DA	Damage Assessment	X	X	X	X
4F9RH	RH Air-Land Repair Equipment	X			
4F9RQ	RH Site Assess Equipment	X			
4F9X1	EOD Core Equipment			X	
4F9X3	EOD Base Support Sustainment Equip.			X	X
4F9X5	EOD Explosive Storage Kit			X	X

UTC	Title	Open	Establish	Operate (Sustain & Recover)	Robust
4F9X8	EOD Demolition Materials			X	X
4F9XN	EOD Agile Combat Employment Support	X	X	X	
4FPRK	RH Air-Land Repair Team	X			
4FPRL	RH Air Land/Insert Repair Team	X			
4FPRQ	RH Site Assess Team	X			
4FPXC	EOD Leadership Support Element			X	X
4FPXD	EOD Core Response Team	X	X	X	X
4FPXG	EOD Flight Leadership Element			X	X
4FWAB	RADR Asphalt Repair Materials		X	X	X
4FWAE	RADR AC Batch Plant		X	X	X
4FWAM	Expeditionary AM-2 Mat Spt. Equip.		X	X	X
4FWCM	RADR Concrete Repair Materials		X	X	X
4FWCR	RADR Crater Repair Equipment		X	X	X
4FWCT	WaFERS-Common Tools & Equip.		X	X	
4FWEA	Expeditionary Airfield Damage Repair			X	
4FWE1	EOD Vehicle, MRAP, Cougar Cat I			X	X
4FWFC	AM-2 Crater FOD Cover	X	X	X	
4FWFD	RADR FOD Removal Vehicles	X	X	X	
4FWFF	WaFERS-Fuel Filter		X	X	
4FWFP	WaFERS-Water and Fuel Pump		X	X	
4FWFR	Fiber Reinforced Polymer Matting	X	X	X	
4FWLH	WaFERS-Lay Flat Hose		X	X	
4FWMS	MAOS Marking Kit	X	X	X	
4FWPS	ADR Paint Striping Kit		X	X	
4FWRB	WaFERS-Repair, Bypass, and Line-Patch		X	X	
4FWSP	Sustainment Pavement Repair Kit		X	X	
4FWSR	Expedient Spall Repair Kit	X	X	X	
4FWTP	WaFERS-Tank Patch		X	X	
4FWWH	RADR Warehouse Vehicles		X	X	
7E1AK	Airborne Airfield Assess Team (CRE)	X			
7E1A3	Airfield Assessment Team (CRE)	X			
7E1A2	Airfield Survey Team (CRE)	X			
9AAB4	PACAF Contingency Response Grp	X			
N/A	**Exp. Trim Pad Anchoring Kit**		X		X

UTC	Title	Open	Establish	Operate (Sustain & Recover)	Robust
N/A	**Exp. Aircraft Tie-down Points**		X		X
UFMAC	20 Ton Tractor 6x4			X	
XFWR4	Mobile Aircraft Arresting System		X	X	
XFWYC	Emergency Airfield Lighting System	X	X	X	
Note: To ensure availability of RADR equipment and expendable repair materials, it must be properly stored and maintained.					

1.2. AFFORGEN Model and Force Elements. The Air Force generates and presents Airmen and airpower to support the Joint Force using the AFFORGEN model (**Figure 1.1**) and Force Elements (**Figure 1.2**) addressed below. **Note:** Force generation and presentation continues to evolve through lessons learned from implementation of AFFORGEN and continued development of operational concepts such as Agile Combat Employment (ACE). Personnel should anticipate changes related to force construct and presentation. See **paragraph 1.2.3** for transitions related to AFFORGEN model and force elements.

1.2.1. The current AFFORGEN model creates a 24-month rotational cycle (Phases of Readiness) with 6-month phases. There are four rotational phases of readiness in the AFFORGEN cycle: Reset (for reintegration and reconstitution), Prepare (for training towards peak readiness), Certify (for certification), Available to Commit (for deployment).

Figure 1.1. AFFORGEN Model.

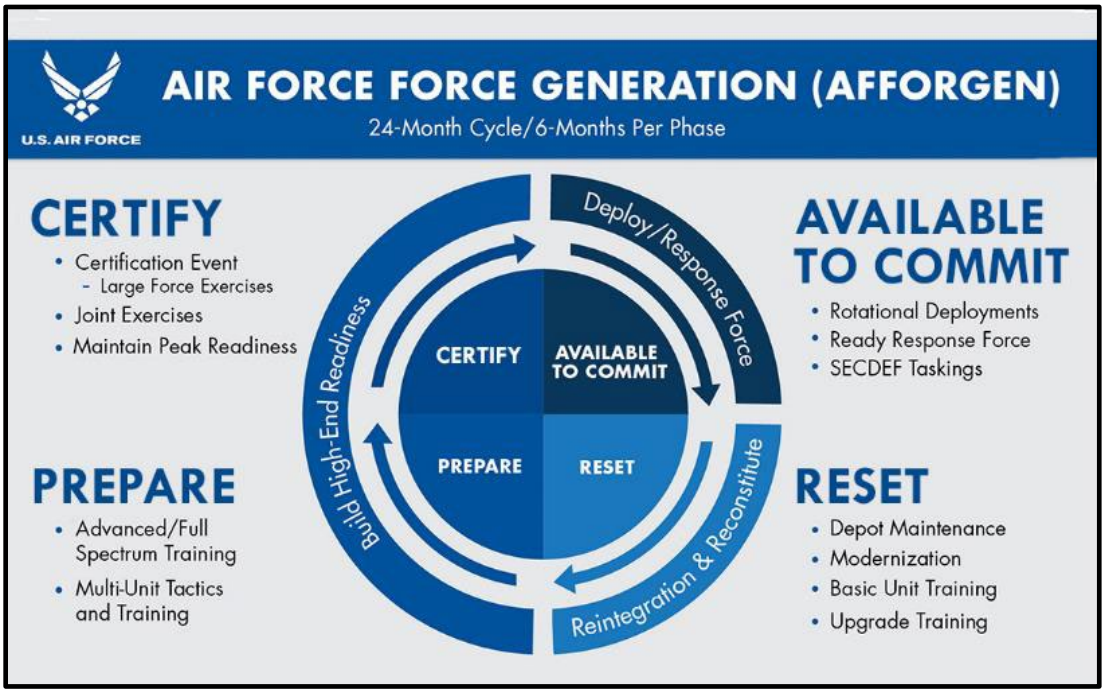
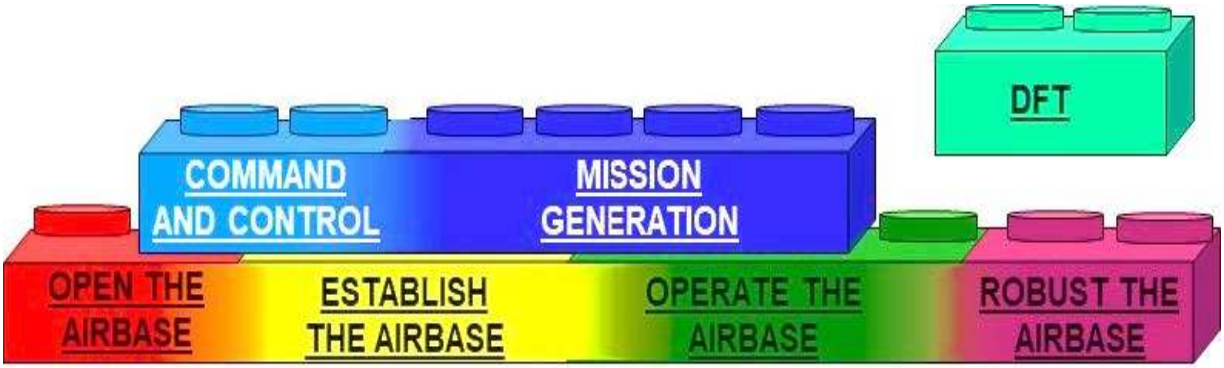


Figure 1.2. AFFORGEN Force Elements.



1.2.2. Under the AFFORGEN model, a Force Element (FE) is an integrated set of UTCs the Air Force uses to offer operational capability to Joint Force Commanders. Joint Force Commanders request AFFORGEN Force Elements through the Global Force Management (GFM) process. **Note:** The default force elements the Air Force offers to Joint Force Commanders to run airbases consist of the Command and Control (C2), Establish the Airbase (EAB), and Operate the Airbase (OAB) force elements. Expeditionary Airbase (XAB) is the GFM-recognized term for these combined force elements. Below is a brief overview of each AFFORGEN Force Element and related civil engineer roles:

1.2.2.1. Open the Airbase FE. Provides capabilities to open an airbase, regardless of follow-on mission. It arrives first and assesses an airbase for establishment of minimum airfield operating parameters. FE sourcing is based on the specific conditions and requirements of individual austere airfields. CE personnel, including RED HORSE and those assigned to Contingency Response Groups, airfield assessment teams, and other small units responsible for specific initial assessment and repair tasks will likely deploy with the Open the Airbase FE. Then, after completing their tasks, redeploy home to reconstitute or forward deploy to open a different site.

1.2.2.2. Command and Control (C2) FE. Provides the capabilities to establish an Air Expeditionary Wing C2 structure to include initial wing, operations, maintenance, mission support, and medical group commanders and staffs. Prime Base Engineer Emergency Force (BEEF) units tasked to beddown and sustain forces at a given site typically deploy a command and control cadre with the Command and Control FE, with the remainder of the tasked Prime BEEF forces deploying with the Establish the Airbase and later FEs.

1.2.2.3. Establish the Airbase (EAB) FE. Provides sufficient forces to bring an airbase to initial operating capability (IOC) status. It contains capabilities to support most missions or weapon systems and will integrate with Open the Airbase and C2 FEs as required. The EAB FE provides the airfield's earliest capability to enable and sustain mission generation. Air Force civil engineers have a significant role when tasked to establish or develop contingency locations supporting Air Force missions and operations, including providing expedient pavement and infrastructure repairs for airfield operations.

1.2.2.4. Operate the Airbase (OAB) FE. Enhances combat support and combat service support capabilities beyond what the EAB FE provides and brings an airbase to full operating capability. OAB enhances force protection, communications, cargo handling, and quality of life activities such as Chaplain Corps, services, health care, and reach-back capabilities. Civil engineers sustain flying operations with day-to-day maintenance of pavement surfaces and, when necessary, rapidly recover airfields after an enemy conventional attack.

1.2.2.5. Mission Generation FE. Provides combat, combat support (those that are specifically identified on the aviation/operational UTC's mission capability statement), and combat service support capability. Mission Generation FEs include Air Superiority, Global Precision Attack, Suppression of Enemy Air Defenses, Long Range Strike, High-Altitude Intelligence, Surveillance, and Reconnaissance, Air Refueling, Intra-Theater Airlift, and Combat Search and Rescue.

1.2.2.6. Robust the Airbase FE. Provides additional combat support and combat service support to increase the robustness of the capabilities already in place through previous FEs. It supports multiple Mission Generation FEs through additive capability. This FE is intended to enable the largest-scale enduring expeditionary operating locations. Civil engineers robust airfield operations with capabilities such as aircraft trim pads, aircraft tie-downs, upgrading expeditionary arresting systems with semi-permanent systems, capping crushed stone repairs with semi-permanent materials to eliminate foreign object debris (FOD) hazards, and expanding aircraft parking ramps and taxiways.

1.2.2.7. Demand Force Teams (DFT). Teams composed primarily of UTCs from units or capabilities that operate independently, including unique, highly specific, combat, combat support, or combat service support capabilities. Example DFT capabilities include but are not limited to the Expeditionary Medical Support System, RED HORSE, and Tactical Air Control Party.

1.2.3. The AFFORGEN model and force elements continue to evolve from lessons learned during implementation. Anticipated changes, in the near-term, may impact force construct and presentation, force elements (FE), UTCs, Mission Capability (MISCAP) statements, and combat service support teams. The expeditionary air base (XAB) will persist as the Air Force transitions to the Air Task Force (ATF) construct.

1.2.3.1. The ATF consists of a Command Element and staff, an Expeditionary Air Base Squadron (EABS) for base operating support; a Mission Generation FE to project air power; and a Mission Sustainment Team (MST) to enable remote operations under the ACE concept of operations.

1.2.3.2. Once implemented, the new ATF construct may continue to evolve and shape force posture and presentation across the spectrum of military operations as we learn and implement best practices. This progression may include significant changes to personnel and equipment UTCs and MISCAPs listed and described in this publication. A good approach is to stay informed on projected changes and maintain flexibility during the transition.

1.3. Tactics, Techniques, and Procedures. The TTPs for ADR and Rapid Airfield Damage Recovery (RADR) continues to evolve, incorporating improved technology, materials, and equipment. As new TTPs are developed and approved, they are incorporated into new or updated AFTTPs. **Table 1.2** lists currently available ADR-related AFTTPs and **Table 1.3** lists additional technical publications associated with ADR operations.

Table 1.2. Airfield Damage Repair Air Force Tactics, Techniques, and Procedures.

ADR Related AFTTPs	
AFTTP 3-32.10	*Introduction to Rapid Airfield Damage Recovery (RADR)*
AFTTP 3-32.11	*Airfield Damage Assessment after Major Attack*
AFTTP 3-32.12	*Minimum Airfield Operating Surface (MAOS) Selection and Repair Quality Criteria (RQC)*
AFTTP 3-32.13	*Airfield Marking and Striping after Major Attack*
AFTTP 3-32.14	*Alternate Installation Sequence for Emergency Airfield Lighting System after Major Attack*
AFTTP 3-32.15	*Rapid Setback Installations of Mobile Arresting Gear System*
AFTTP 3-32.16	*Sustaining Airfield Pavement at Enduring Contingency Locations*
AFTTP 3-32.17	*Rapid Airfield Crater and Spall Repair*
AFTTP 3-32.18	*Rapid Airfield Damage Repair Warehouse Operations*
AFTTP 3-32.19	*Rapid Airfield Damage Repair Asphalt Batch Plant Operations*
AFTTP 3-32.22	*Water and Fuel Expedient Repair System (WaFERS)*
AFTTP 3-32.5V6	*Explosive Ordnance Disposal (EOD) Unexploded Explosive Ordnance (UXO) Response Operations*

Table 1.3. Additional Airfield Damage Repair Technical Publications.

ADR Technical References	
T.O. 35E2-2-7-WA-1	*AM-2 Airfield Landing Mat and Accessories*
T.O. 35E2-5-1-WA-1	*Crushed Stone Crater Repair and Line-of-Sight Profile Measurement for Rapid Runway Repair*
TSPWG M 3-270-01.3-270-07	*Airfield Damage Repair*
UFC 3-250-03	*Standard Practice Manual for Flexible Pavements*
UFC 3-250-08FA	*Standard Practice for Sealing Joints and Cracks in Rigid and Flexible Pavements*
UFC 3-250-09FA	*Aggregate Surfaced Roads and Airfields Areas*
UFC 3-250-11	*Soil Stabilization and Modification for Pavements*
UFC 3-260-02	*Pavement Design for Airfields*
UFC 3-260-03	*Standard Practice for Airfield Pavement Evaluation*

ADR Technical References	
UFC 3-260-04	*Airfield and Heliport Marking*
UFC 3-260-11FA	*Model Design-Build (D-B) Request for Proposal (RFP) For Airfield Contracts*
UFC 3-260-16	*O&M Manual: Standard Practice for Airfield Pavement Condition Surveys*
UFC 3-260-17	*Dust Control for Roads, Airfields and Adjacent Areas*
UFC 3-270-01	*O&M Manual: Asphalt and Concrete Pavement Maintenance and Repair*

Chapter 2

AIRFIELD DAMAGE REPAIR (ADR) UNIT TYPE CODES (UTC)

2.1. Overview. This chapter provides engineering, operations, and logistics planners with ADR capability (UTC) descriptions. Most ADR UTCs are not paired with dedicated personnel UTCs; ADR tasks are performed by in-place forces (either permanent party or beddown forces) and augmented with deployed forces when necessary. *The Civil Engineer Supplement to the War and Mobilization Plan-Volume 1* (WMP-1) lists and describes all CE personnel and equipment UTCs. Because there are many combinations of CE personnel UTCs capable of performing ADR tasks, Prime BEEF personnel UTCs will not be discussed within this pamphlet; however RED HORSE, Contingency Response forces, and Explosive Ordnance Disposal personnel UTCs are discussed where applicable.

2.2. 4F9AD—Airfield Pavements Evaluation Team. Utilizes destructive and non-destructive techniques to evaluate pavement structural integrity (**Figure 2.1**). Provides proof loading of aircraft anchoring systems, friction characteristics testing, technical expertise and guidance for the design, construction and repair of airfield pavements in support of regional conflict operations at contingency operating locations, aerial ports, enroute bases, or critical stateside bases to include humanitarian relief operations. Contact the Air Force Civil Engineer Center (AFCEC)Reachback Center (850-283-6995, DSN 523-6995 or AFCEC.RBC@us.af.mil) for more information.

Figure 2.1. Conducting Airfield Pavement Evaluation.



2.3. 4F9DA—Damage Assessment. Provides remote sensing and geospatial information system technologies to perform airfield damage assessment to recover an airfield after attack. Personnel requirements are identified in AFTTP 3-32.12 and must be sourced from the following personnel beddown UTCs when in-place forces are not available: 4FPET, 4FPAV, and 4FPXC.

2.4. 4F9RH—RED HORSE Repair Team Equipment. Specialized air insertable equipment UTC that gives air land/sling load RED HORSE the capability to make expedient airfield damage repairs and provide a quick assessment of follow-on deployment needs. Provides equipment to perform initial site survey assessment (GeoReach); demolition of obstructions; expedient force

protection construction; airfield surface repairs for C-130/C-17 operations; EALS installation; potable water source testing; and pavement evaluations. This UTC is linked with **4FPRL**.

2.5. 4F9RQ—RED HORSE Site Assessment Equipment. This UTC provides equipment for site assessment engineer force to perform advanced airfield survey, site layout, and planning for establishment and future development of an operating location during contingencies.

2.6. 4F9X1—EOD Core Equipment. Specialized equipment and explosives required in the initial EOD support of major combat, contingency, and homeland operations. Equipment and explosives are loaded on vehicles to ensure EOD response teams arrive with and can employ all required equipment and personnel from a single aircraft. Provides limited capability to assess and mitigate chemical, biological, radiological, and nuclear (CBRN) threats. UTC must be paired with (1) UFM3X, Up-armored high mobility multi-purpose wheeled vehicle, (1) UFMTB, Hi-mobility Trailer, vehicle UTCs, and (1) **4FPXD** personnel UTC to be fully mission capable. Vehicle support is paramount to this capability. UTC contains sensitive/mission critical EOD equipment requiring an EOD escort (**4FPXC** or 4FPXD UTC) during transport.

2.7. 4F9X3—EOD Base Support Sustainment Equipment. EOD specialized response team equipment and explosives required to provide base sustainment in support of major combat, contingency, and homeland operations where EOD Core UTCs (**4F9X1**) already exist or are scheduled for deployment. Protects personnel, and resources, from the effects of explosive hazards, munitions, accidents and unexploded ordnance to include improvised explosive devices. Provides limited capability to assess, disrupt, neutralize or render safe chemical, biological, radiological, nuclear, and explosive threats. Contain sensitive/mission critical EOD equipment requiring an EOD escort (**4FPXC** or **4FPXD**) UTC) during transport.

2.8. 4F9X5—EOD Explosive Storage Kit. Deployable explosive storage magazine set used to store EOD demolition packages required in support of major combat, contingency, and homeland operations. Magazine designed to significantly reduce explosive hazard class and quantity distance requirements used for temporary explosive storage during initial contingency operations until siting of permanent explosive storage areas are established. Magazines employed during homeland operations to support EOD emergency responses. UTC includes Air Force standard security system required for storage of high risk munitions.

2.9. 4F9X8—EOD Demolition Materials. Provides EOD demolition materials for response teams clearing multiple unexploded ordnance (UXO) during rapid airfield damage recovery operations. UTC can be utilized to resupply teams with explosives necessary to support explosive ordnance disposal teams' conducting joint sourced or operations in high threat areas where multiple explosive hazards must be cleared.

2.10. 4F9XN-EOD Agile Combat Employment Support. Provides force protection and tactical aircraft clearing tools required to support combatant command assets, tactical aircraft and personnel, in austere locations. Provides 24-hour support to render safe all tactical aircraft weapons systems and munitions incidents as well as provide force protection capabilities to detect and defeat terrorist devices.

2.11. 4FPRL (Active Duty)/4FPRK (Air Reserve Component)—RED HORSE Air Insert Engineer Repair Team. This 21-person UTC Provides capability for Air-Insertable/Air-Assault team delivered via air-land, or air-insert into bare base or forward operating locations. Provides initial site survey assessment (GeoReach); removal/demolition of obstructions; expedient force protection construction; repair of airfield surfaces for limited C-130/C-17 operations; installation of Emergency Airfield Lighting System (EALS); testing for potable water sources; and pavement evaluations. Air-Insertable RED HORSE flight and supporting forces are organized and equipped to provide initial operational capability for 14 days maximum before follow-on RED HORSE 4FPR* series UTCs are required.

2.12. 4FPRQ—RED HORSE Site Assessment Team. This 12-person site assessment engineer force performs advanced airfield survey, site layout, and planning for establishment and future development of a contingency operating location during contingencies (**Figure 2.2**). The team is self-sustaining for five days, and longer if resupplied. Contains site development, field engineering, and airfield pavement evaluation capabilities. Requires UTC **4F9RQ** to provide site assessment equipment for full capability.

Figure 2.2. Airman Going Out for Site Assessment.



2.13. 4FPXC—EOD Leadership Support Element. Basic Engineer Force to beddown, sustain, and recover garrison and contingency locations for natural or manmade disaster response. EOD leader/support enablers required in support of major combat, contingency, and homeland defense operations. For full requirements see CE Supplement to the WMP-1.

2.14. 4FPXD—Explosive Ordnance Disposal (EOD) Core/Response Team. Three person EOD team required in support of major combat, contingency, and homeland defense operations (**Figure 2.3**). UTC is linked with **4F9X1** or can augment existing requirements. Combat support is required from service designated as base operating support-integrator. Personnel deploy with individual protective equipment, clothing, weapons (M4 and M9) and ammunition. One 7-level TSgt and one 5-level required as a minimum; third person may be a 3E831. See T.O. 35E8-2-10-1, *Arresting Systems, Aircraft, Mobile*, and AFTTP 3-32.15, *Rapid Setback Installation of Mobile Arresting Gear System*, for installation and operating procedures.

Figure 2.3. EOD Technician Probing for Unexploded Explosive Ordnance (UXO).



2.15. 4FPXG—EOD Flight Leadership Element. Basic Engineer Force to beddown, sustain, and recover garrison and contingency locations for natural or manmade disaster response. EOD C2 team required to support major combat, contingency, and homeland defense operations. Provides logistical support, trends, and incident reporting analysis, and standardized guidance for operations. Long-term or broad scale operations may require a multi-disciplined support element for continued support. For full requirements, see CE Supplement to the WMP-1.

2.16. 4FWAB—RADR Asphalt Repair Materials. This UTC provides certified semi-permanent crater repair material (flowable-fill and asphalt) to rapidly repair airfield pavement damage, usually after an attack (**Figure 2.4**). Contains material quantities for 18 repairs approximately 8.5-ft by 8.5-ft x 2-ft.

Figure 2.4. Airmen Finishing Asphalt Cap Repair Material.



2.17. 4FWAE—RADR AC Batch Plant. This UTC provides vehicles/equipment to operate an asphalt batch plant (**Figure 2.5**). Plant (two recyclers) can produce up to 14 tons per hour. See AFTTP 3-32.19, *Rapid Airfield Damage Repair Asphalt Batch Plant Operations*, for information on RADR Asphalt Batch Plant operations.

Figure 2.5. Airmen Batching Asphalt for Crater Repairs.



2.18. 4FWAM—Expansion AM-2 Mat and Support Equipment. Provides Aluminum Matting, Version 2 (AM-2), for the installation of a standard 216-ft x 198-ft (42,768-sf) aircraft pad to support parking, taxiway expansion (72-ft x 200-ft), or repair on asphalt, concrete or stabilized ground (**Figure 2.6**).

Figure 2.6. Engineers Rapidly Expanding Aircraft Parking Ramp with AM-2.



2.19. 4FWCM—RADR Concrete Repair Materials. This UTC provides certified semi-permanent crater repair materials (flowable-fill and rapid-set concrete) for rapid crater repair (**Figure 2.7**). Contains material quantities for 18 repairs approximately 8.5-ft x 8.5-ft x 2-ft in depth or two repairs 30-ft x 30-ft x 2-ft in depth.

2.20. 4FWCR—RADR Crater Repair Equipment. This UTC provides the ability to rapidly repair craters up to 30 foot in diameter (**Figure 2.8**). Multiple UTCs will be required based upon battle damage assessment or directive; single UTC equates to "small" capability (repairs 18 small [8.5-ft x 8.5-ft x 2-ft] or 2 large [30-ft x 30-ft x 2-ft] craters within 6.5 hours), three (3) each UTCs equates to "medium" capability (54 small or 6 large craters), five (5) each UTCs equates to "large" capability (90 small or 10 large craters) and seven (7) each UTCs equates to "very large" (126 small or 14 large craters).

Figure 2.7. Airmen Using Screed to Finish Rapid-Set Capping Repair Material.



Figure 2.8. Crater Repair Vehicles in Action.



2.21. Water and Fuel Expedient Repair System (WaFERS). WaFERS encompasses capabilities required to repair and recover water or fuel infrastructure at U.S. or allied airfields. Capabilities include, but not limited to, pipe repair or bypass; valves/manifold repair or bypass; hydrant capping; storage tank patching; fuel or water pumping; fuel filtering, and fuel storage. The UTCs may standalone or conjoined to restore water or fuel system operations. The modularity and scalability of WaFERS allows UTC selection based on the installation's unique infrastructure. Input and output connection points for all components are standard DOD configuration allowing for the use of quick-connect type fittings and couplers to the maximum extent possible. See AFTTP 3-32.22, *Water and Fuel Expedient Repair System (WaFERS)*, for more information. WaFERS consists of seven Unit Type Codes and are summarized below. **Note:** The following WaFERS UTCs are currently in the process of being registered and were not finalized before this was published.

2.21.1. 4FWCT—Common Tools and Equipment. Provides necessary tools and equipment to support expedient repair and recovery of water or fuel distribution and storage systems damaged by attack, natural disaster, or catastrophic failure (**Figure 2.9**). These tools are likely to be used with all WaFERS UTCs to accomplish any repair.

Figure 2.9. Examples of WaFERS Consolidated Tools and Equipment.



2.21.2. 4FWRB—Repair, Bypass, and Line-Patch. The Repair, Bypass, and Line-Patch UTC offers several capabilities to include repairing large leaks or punctures in a pipeline, fitting or valve; inline repair; valve replacement; and hydrant capping (**Figure 2.10**). Bypassing is defined as rerouting the flow through a hose or pipe in order bypass a damaged system or to go around a damaged area due to hazards, traffic patterns and paths or congestion. The Line Patch can repair small leaks or punctures in a pipeline or fittings such as elbows, valves, or reducers without replacing large sections of the pipeline or fitting. Repair material comes in the form of epoxies, polyester wrap, wood plugs, and clamps of various types and pressure ratings to address many types of damage (**Figure 2.11**). When this UTC is combined with other WaFERS UTCs (such as the pump and filter UTCs), the capability is greatly enhanced and expanded including bypassing a pump house or manifold to get fuel to a hydrant loop system.

Figure 2.10. A Bypass Repair.



Figure 2.11. Airmen Performing Line Patch Repair.



2.21.3. 4FWLH—Lay Flat Hose. Lay flat hose can be used to quickly replace significant sections of above grade pipe that requires replacement or to bypass damaged or inoperable infrastructure such as a pump house (**Figure 2.12**). This hose can be cut to length as needed and has field assembly capable hose ends. Hoses are available in 6-inch, 8-inch, and 10-inch diameters and can run up to 600 feet. Twelve-inch diameter hoses are available in shorter segments in the Repair and Bypass UTC. The pressure rating on the hoses is 150-PSI and can be used up to ANSI 150 class pressures when absolutely required; however, the safety factor is reduced.

Figure 2.12. Repairing Piping with Lay Flat Hose.



2.21.4. 4FWTP—Tank Patch. This UTC has the capability to repair a leak due to a penetrating hole, puncture or small tear in the wall of a bulk fuel storage tank. This capability has several methods of repair to choose from including wooden plugs driven into the leaking hole with a hammer and the umbrella patch which is to be used on larger penetrations (**Figure 2.13**). Depending on what method is chosen, a leak the size of a pinhole and up to 23 inches in diameter can be repaired.

Figure 2.13. Installed Tank Patch.



2.21.5. 4FWFP—Water and Fuel Pump. The Water and Fuel Pumping UTC consists of a trailer mounted diesel engine driven pump capable of pumping liquid from approximately 50 foot below grade with discharge flow of 600-GPM and an outlet pressure of 150-PSI (**Figure 2.14**). This capability works in conjunction with the Fuel Filtering UTC. This pump is capable of series (higher pressure) and parallel (additional flow rate) configurations. Approved pump engine fuels are JP-5, JP-8, Jet-A, kerosene, and diesel fuel which can be scavenged from the pumped flow to fuel the engine. A dedicated hydraulic power unit is used for hydraulic tooling and for extracting fuel out of cut and cover fuel tanks by the down-hole turbine pump. When combined with the lay flat hose UTC, it can be used in series or parallel to meet flow or pressure demands of a hydrant loop or other fuel systems.

Figure 2.14. Water and Fuel Pump with Hoses Connected.



2.21.6. 4FWFF—Fuel Filter. The Fuel Filtering UTC consists of a trailer mounted filter system with two 600-GPM filter separators (**Figure 2.15**). This capability comes with numerous hose connection adapters to facilitate connection to the Fuels Operational Readiness Capability Equipment or other systems providing lateral support capability. This UTC provides expedient filtering of stored or recovered fuel from pipelines, berms, craters, and storage tanks damaged by attack, natural disaster, or catastrophic failure at main or contingency operating locations, aerial ports, enroute bases, and joint bases. Fuel filtering provides suitable advanced filtration for contaminated fuels reintroduced into fuel storage or distribution systems.

Figure 2.15. Trailer Mounted Fuel Filter System.



2.22. 4FWE1—EOD Vehicle, MRAP, Cougar Cat I. Provides initial WRM EOD Mine Resistant Ambush Protection Vehicle (MRAP) Cougar Cat I (Configure with associated Mission Support Equipment [MSE]) **(Figure 2.16)** . The following MSE is general in nature: Communications, Blue Force Tracker, Counter Radio-Controlled IED Electronic Warfare (CREW), off road survival tool and miscellaneous support equipment. UTC requires qualified vehicle mechanic support (UFMX4). Mounted spare tire, applicable TOs, operation/repair manuals and Temporary Mission Support Kits will accompany vehicle. Specific MSE required for the EOD mission sets is identified in the Logistics Detail. No suitable MRAP substitution are authorized. For additional assets and for follow-on EOD teams use the EOD MRAP mobility UTC.

Figure 2.16. MRAP Cougar.



Note: The EOD MRAP mobility UTCs were in development and had not been registered at the time of this writing. Provides roll-on/roll-off capability for EOD teams with critical personal protective equipment required in support of Counter-IED route clearance, base recovery, force protection, render safe procedures and/or weapons of mass destruction operations in high, medium, and low threat areas. Can be employed in the support of major combat, contingency, and homeland defense operations.

2.23. 4FWEA—Expeditionary Airfield Damage Repair. Provides equipment, material, and vehicles for the recovery of airfield pavements damaged after attack or natural disaster. The capability includes the ability to repair 100 spalls, supports limited inclement weather capabilities, and provides limited foreign object debris (FOD) removal. To be air-liftable by C-130 or similar aircraft, must tailor out the excavator (with attachments). Must deploy with 2x 4FWFR to be fully mission capable. For full FOD capability, must deploy with 4FWFD. Manning requirements are identified in the WMP-1 CE supplement. Supports Operate the Airbase (including airfield recovery after attack) Force Element.

2.24. 4FWFC—AM-2 Crater FOD Cover. Facilitates rapid restoration and reopening of airfield operating surfaces after attack (**Figure 2.17**). Provides AM-2 matting, specialized components and tools to assemble and cover thirty-one 10-ft repairs, nine 15-ft repairs, or three 30-ft repairs. This UTC supports repairs on asphalt, concrete, or compacted aggregates/soils.

Figure 2.17. Airmen Constructing an AM-2 FOD Cover.



2.25. 4FWFD—RADR FOD Removal Vehicles. This UTC provides the vehicles, such as vacuum sweepers (**Figure 2.18**), to clear debris from the airfield after an attack.

Figure 2.18. Vacuum Sweeper Removing FOD from MAOS.



2.26. 4FWFR—Fiber Reinforced Polymer (FRP) Matting FOD Cover. Facilitates rapid restoration and reopening of airfield operating surfaces after attack. Provides FRP matting, specialized components, and tools to assemble and cover twenty 10-ft craters, three 30-ft craters, or one 50-ft crater. The kit includes 35 full-size panels, 30 half-size panels, 15 full-size anchor panels, 20 half-size anchor panels, 760 lower joining bushings, 875 upper joining bushings, and 625 anchor bushings. This UTC supports repairs on asphalt or Portland cement surfaces (**Figure 2.19**). **Note:** Replaces legacy Folded Fiberglass Matting.

Figure 2.19. Airmen Constructing FRP FOD Cover.



2.27. 4FWMS—Minimum Airfield Operating Surface (MAOS) Marking Kit. This UTC provides threshold and departure end markers, edge markers, distance-to-go markers, and aircraft arresting system markers to provide visual clues for takeoff and landing surface (**Figure 2.20**). See T.O. 35E2-6-1, *Minimum Operating Strip Marking System, Layout and Marking Procedures for Rapid Runway Repair*, and AFTTP 3-32.13 for installation and operating procedures. **Note:** This UTC is in development and had not been registered prior to this being published.

Figure 2.20. MAOS Marking System Components.



2.28. 4FWPS—ADR Paint Striping Kit. Facilitates the rapid restoration and opening of airfield operating surfaces after attack. Provides ability to stripe a MOS up to 50-ft x 150-ft wide and up to 10,000-ft long, and taxiway turn- and hold-lines (**Figure 2.21**). The kit also provides blackout capability of unwanted stripes and turn-lines to prevent pilot confusion. See T.O. 35E2-6-1 and AFTTP 3-32.13 for more information on airfield striping.

Figure 2.21. Paint Striping Machine Example.



2.29. 4FWSP—Sustainment, Pavement Repair (SuPR) Kit. Provides initial WRM equipment/ vehicle capability for minor airfield pavement repairs such as spalling, small potholes, joint repair; the kit is not tailored for crater repairs (**Figure 2.22**). Kit contains multi-use equipment and materials to support Prime BEEF, RED HORSE, and/or other civil engineer units operating in austere locations. Kit also contains specialized equipment and certified materials to maintain and repair both asphalt-based and/or Portland cement-based pavements with semi-permanent materials to include limited joint sealing. See AFTTP 3-32.16 for information on this pavement repair kit.

Figure 2.22. Views of SuPR Kit Contents.



2.30. 4FWSR—Expedient Spall Repair Kit. Provides tools and materials to expeditiously perform temporary repairs on up to 200 spalls approximately 24 inches in diameter and 6 inches deep (**Figure 2.23**). Requires UFMBG with UFMAQ or UFMBP to transport personnel, equipment, and material. Must task one UTC per **4FWCR** tasked.

Figure 2.23. Airman Performing Expedient Spall Repair.



2.31. 4FWWH—RADR Warehouse Vehicles. This UTC provides material handling vehicles to load and deliver repair materials to the airfield (**Figure 2.24**). See AFTTP 3-32.18 for information on RADR Warehouse operations.

Figure 2.24. Warehouse Vehicles Delivering & Unloading Repair Materials at Repair Zone.



2.32. 7E1AK—Airborne Airfield Assessment Team. A 15-person team developed from a Contingency Response Squadron providing advanced echelon, airfield/ landing zone survey and assessment, site planning, and airfield opening (**Figure 2.25**). Determines airfield suitability to support flying operations to include command, control, communication, air traffic control, weather, airfield pavements and obstruction criteria, logistics support, infrastructure, force protection, initial beddown and overall airfield support. Capable of fixed-and-rotary-wing air land or airborne (static-line) infiltration into permissive or uncertain environments. Self-sustaining for five days. One qualified airfield survey team (**7E1AP**) required to certify surveys.

Figure 2.25. Airborne Airfield Assessment Team Preparing for Airfield Survey.



2.33. 7E1A3—Airfield Assessment Team. Includes equipment to conduct airfield assessments/surveys in accordance with AFI 10-202, *Contingency Response Forces*, in permissive to uncertain threat environments and determines airfield suitability for all Air Mobility Command aircraft operations. Team from Contingency Response Group is capable of self-sustained operations for five days to assess/report status of organic command and control (C2), communications, air traffic control, airfield pavements, infrastructure, force protection, and overall airfield condition to combatant commander to determine suitability for mobility/combat operations. Must be tasked with 7E1A2 for personnel.

2.34. 7E1A2—Airfield Survey Team. Contingency Response Group/Contingency Response Squadron eight-person team to conduct worldwide airfield and site surveys to assess airfield capabilities in support of air mobility operations. Requires expeditionary combat support. Additional support will be required if deployed to a bare base.

2.35. 9AAB4—PACAF Contingency Response Group. Provides 21 Group personnel and equipment to augment PACAF's "first-in" contingency response element to establish initial site/airfield/airbase operations in the Pacific Command Theater. Capable of performing a wide spectrum of contingency operations including base opening, noncombatant evacuation operations, humanitarian assistance/disaster response, and airfield seizure liaison. Core capabilities include 12-hour response to assess/manage/provide: engineering, bare base planning, pavement evaluation, facilities, airfield, air traffic control, communications, limited force protection, site security, surveillance, limited CBRN assessment/detection, aircrew flight equipment, logistics, supply, knowledge operations, client systems technology, and initial Personnel Support for Contingency Operations. Able to operate in permissive and uncertain threat environment, and in low/no light conditions. Must be augmented by 7E1BD for beddown and 5-day sustainment. This is not a stand-alone UTC; must be tasked in conjunction with airbase opening UTCs: 7E1AE, 7E1BD, 7E1CA, QFEPF, UFBBR. Must be relieved/augmented within 45 days. Provides multi-skilled 12-hour response airfield/site assessment team to determine suitability of airfield/sites.

2.36. Expedient Trim Pad Anchoring and Contingency Aircraft Tie-Down Anchors. These capabilities are not provided by UTCs.

2.36.1. Expedient Trim Pad. The expedient trim pad is used when the construction of a permanent trim pad is not practical. There are two types of anchor systems for the expedient trim pad: Earth Anchor System and Steel Plate and Rod System.

2.36.1.1. Earth Anchor System. 12-stake anchor group (**Figure 2.26**) constructed in soil. Supports a maximum loading of approximately 40,000 lbs (177.9 Kilo Newtons). All hardware and tools required for installation (except the restraining cable) is sourced from the Mobile Aircraft Arresting System UTC (XFAR4).

2.36.1.2. Steel Plate and Rod System. Sited in a sound concrete slab with a thickness of at least 12 in (305 millimeters) (**Figure 2.27**). Supports a maximum loading of approximately 70,000 lbs (311.4 Kilo Newtons).

Figure 2.26. 12-Stake Earth Anchor System.

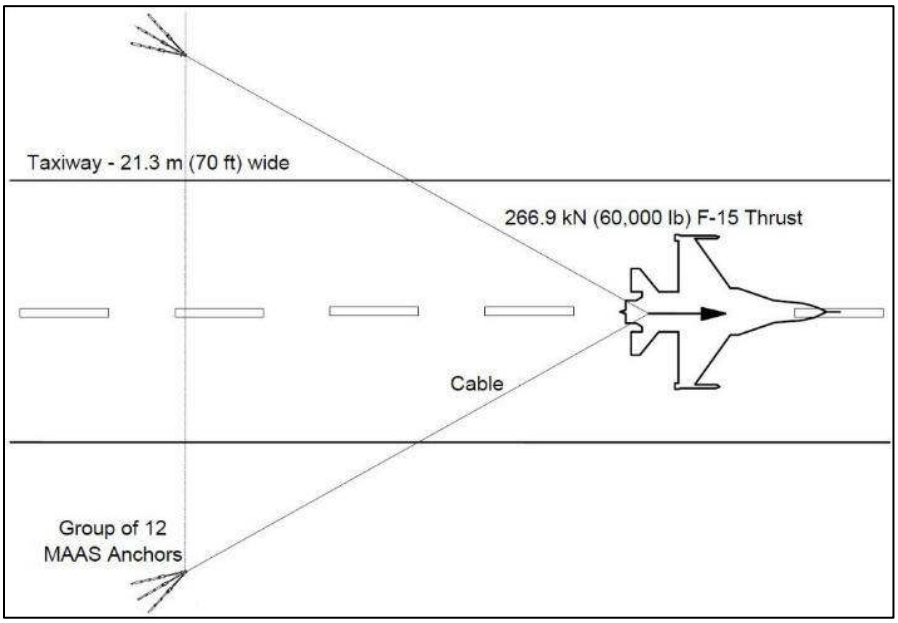
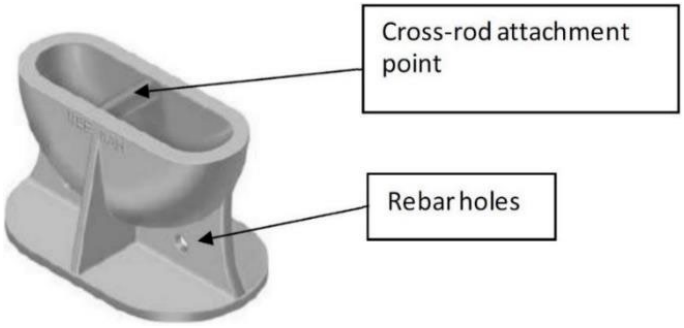


Figure 2.27. Steel Plate and Rod Trim Pad System.



2.36.2. Contingency Aircraft Tie-Downs. Aircraft tie-downs are an integral component of airfield aprons (**Figure 2.28**). An aircraft must be tied to a tie-down to ensure it remains stationary in the event of various weather phenomena such as high winds. Additionally, activities such as maintenance and loading require that aircraft remain stationary. In some instances, tie-downs also serve as static grounding points; discussion herein focuses on tie-downs specifically designed to physically secure aircraft.

Figure 2.28. Neenah Mooring Eye.



2.36.2.1. The current guidelines characterize aircraft tie-downs by the following criteria: lightweight tie-downs are capable of an ultimate uplift resistance of 17,000 lbs (7711 kilograms) and heavyweight tie-downs are capable of an ultimate uplift resistance of 37,700 lbs (17,100 kilograms). Testing conducted by the Air Force Research Laboratory determined that two tie-down systems meet the lightweight criteria, and one system meets the heavyweight tie-down criteria.

2.36.2.2. Additional information regarding tie-down spacing and layout information is available in UFC 3-260-01, *Airfield and Heliport Planning and Design*.

2.37. UFMAC—20 Ton Tractor 6 x 4. B364 vehicle only (**Figure 2.29**). Provides using unit initial base operating support capability for cargo movements. Requires qualified vehicle mechanic support (this UTC supports movement of RADR trailers from staging area to the airfield during RADR team mobilization).

Figure 2.29. 20-Ton Tractor 6 x 4.



2.38. XFWR4—Mobile Aircraft Arresting System. Provides austere base with bi-directional arrests of aircraft weighing up to 58,000 lbs at 180 knots on runways at least 1,200-ft long and up to 150-ft wide (**Figure 2.30**). Recycle capability is 20 arrests per hour. Includes one set of Mobile Aircraft Arresting System trailers (two trailers) and two Mobile Runway Edge Sheaves. Must be tasked to support beddown of combat Air Forces where no in-place arresting system exists. Barrier maintenance UTC XFSCL or XFACL is required to expedite system installation. See T.O. 35E8-2-10-1 and AFTTP 3-32.15 for installation and operating procedures.

Figure 2.30. Mobile Aircraft Arresting System Installed with Mobile Runway Edge Sheave.



2.39. XFWYC—Emergency Airfield Lighting. Provides austere base with Emergency Airfield Lighting System (EALS) to light airfields from a 5,000 foot by 50 foot minimum operating strip (MOS) up to a 10,000 foot by 150 foot runway (**Figure 2.31**). Includes distance-to-go markers, runway edge lights, taxiway lights, strobe lights for approach, threshold lights, and precision approach path indicators. Also includes two MEP-805 generators to power the system. See T.O. 35F5-3-17-1, *Lighting System, Airfield, Emergency AE82U-2, 99119653-1 (EALS-A), 88119663-3 (EALS-B)*, for installation and operating instructions and AFTTP 3-32.14, *Alternate Installation Sequence for Emergency Airfield Lighting System after Major Attack,* for alternate installation sequence after a major attack.

Figure 2.31. Airmen Installing EALS Strobe Segment.



Chapter 3

OPEN THE AIRBASE CAPABILITIES

3.1. Overview. Initial capabilities in the Open the Airbase force element (FE) consists of command and control (C2), force protection, cargo and passenger handling, logistics, airfield operations, force accountability, finance and contracting, host nation support, reception, and beddown of follow-on forces. Employment of this FE is necessary when airbase infrastructure and conditions do not meet the minimum requirements of the Establish the Airbase FE.

3.1.1. Early in the planning process, planners from operations, logistics, and engineering identify potential airfields to support the scheme of operations and logistics buildup. Existing imagery, airfield and facility surveys are referenced if available. Planning considerations include siting within the operational area, explosive hazards, Petroleum, Oil, and Lubricants, terminal control, potable water, potential public health hazards, and other environmental issues.

3.1.2. In semi-permissive environments, the 554th RED HORSE Squadron (with their mobile airfield repair equipment set) may arrive via ground movement, air-inserted, or air-delivered in support of an Air Mobility Command Contingency Response Group. These forces conduct airfield assessments, establish a MOS, and begin basic preparations for follow-on forces.

3.1.3. Initial focus is to establish the MOS with sufficient length, width, and weight bearing capacity to support the first mobility missions. This “opening” force enables reception of follow-on engineer forces and may redeploy upon arrival of follow-on capability. Not all UTCs listed in this chapter may be required for every “opening” mission; task UTCs as needed.

3.2. Capabilities. The capabilities contained within this force element (FE) should rapidly establish an IOC in approximately 24 hours from the arrival of forces. The primary constraint is the "Establish Minimum Operating Strip" capability. If extensive runway preparation is required, IOC for opening the airbase is approximately 36 hours. The following ADR capabilities accomplish specific airbase opening processes.

3.2.1. Assess the Airbase. Open the Airbase forces will normally arrive first and assess the airbase to establish minimum airfield operating parameters, command and control, and supporting host-nation support capabilities. The ADR associated UTCs that provide “assess the airbase” capability are identified in **Table 3.1**.

Table 3.1. ADR Assess the Airbase Capabilities.

UTC	Title	PAX	Short Tons
4FPRQ	RED HORSE Site Assessment Team	12	
4F9RQ	RED HORSE Site Assessment Equipment		13.6
4F9AD	Airfield Pavements Evaluation Team	4	27.7
4F9DA	Damage Assessment		0.2

4FPXD	EOD Core Response Team	3	0
4F9XN	EOD Agile Combat Employment Support	0	2
7E1AK	Airborne Airfield Assess Team	15	6.3
7E1A3	Airfield Assessment Team	8	9.9
7E1A2	Airfield Survey Team	8	0.1
9AAB4	PACAF Contingency Response Group	21	49.4
Note: The PAX and short tons listed above are a template. These may vary due to individual unit LOGFORs. Units should verify PAX and weight requirements in DCAPES or the WMP-1 https://usaf.dps.mil/sites/13072/SitePages/Publications.aspx			

3.2.2. Establish Minimum Operating Strip (MOS). Open the Airbase engineer forces perform expeditionary airfield pavement repairs to establish a MOS for C-130 and C-17 aircraft. Repair assets and capabilities may be limited until minimum repairs are completed to enable cargo aircraft to bring additional resources to establish the airbase. In a non-permissive environment, choose the applicable RED HORSE repair team in **Table 3.2** to marry with the **4F9RH** equipment set to establish the MOS. Prime BEEF engineers typically operate in permissive environments.

Table 3.2. ADR Capabilities to Establish a MOS.

UTC	Title	PAX	Short Tons
4FPRK	RED HORSE Air Insert Engineer Repair Team (Air Reserve Component)	21	
4FPRL	RED HORSE Air Insert Engineer Repair Team	21	
4F9RH	RED HORSE Air Insert Engineer Repair Team Equip.		47.3
4FWFR	FRP Matting		154.5
4FWMS	MAOS Marking Kit		*4.7
4FWFC	AM-2 Crater FOD Cover		43.4
4FWSR	Expedient Spall Repair Kit		11.5
4FWFD	RADR FOD Removal Vehicles		123.5
4F9X8	EOD Demolition Materials		*5.8
*Under development. Note: The PAX and short tons listed above are a template. These may vary due to individual unit LOGFORs. Units should verify PAX and weight requirements in DCAPES or the WMP-1 https://usaf.dps.mil/sites/13072/SitePages/Publications.aspx			

Chapter 4

ESTABLISH THE AIRBASE CAPABILITIES

4.1. Overview. The Establish the Airbase (EAB) force element (FE) brings the base to an IOC within four days of arrival to support most missions or weapon systems.

4.1.1. Follow-on ADR resources within this FE include Prime BEEF personnel, vehicles, heavy equipment with work tool attachments, and specialized toolkits/materials to make repairs and upgrades to airfield infrastructure, improve repairs made by "Open the Airbase" engineer forces, and to repair more extensive damage and/or expand airfield operating surfaces (ramps and taxiways).

4.1.2. Additional capabilities within the EAB FE include mitigating explosive hazards, airfield lighting, aircraft arresting system, airfield fuel systems repair, and airfield pavement marking and striping. These supporting systems may require additional, or specialized, resources for installation, and maintenance. Engineers maximize the use of local materials to minimize the logistics footprint. As necessary, airfield infrastructure is expanded to improve airbase resiliency and repairs are continued towards meeting full operating capability.

4.2. Capabilities. ADR Capabilities within this FE are identified in **Table 4.1**.

Table 4.1. ADR Capabilities to Establish the Airfield.

UTC	Title	PAX	Short Tons
XFWYC	Emergency Airfield Lighting System		*
4FPXD	EOD Core Response Team	3	*
4F9XN	EOD Agile Combat Employment Support		2
4F9X8	EOD Demolition Materials		*5.8
4F9DA	Damage Assessment		0.2
4FWSP	Sustainment Pavement Repair Kit		*
4FWFR	Fiber Reinforced Polymer (FRP) Matting		23.5
4FWMS	MAOS Marking Kit		*
4FWAM	WRM Expansion AM-2 Mat & Support Equip		174.4
XFWR4	Mobile Aircraft Arresting Systems		*
4FWPS	ADR Paint Striping Kit		9.4
4FWSR	Expedient Spall Repair Kit		9.4
4FWFD	RADR FOD Removal Vehicles		123.6
4FWCT	WaFERS-Common Tools & Equipment		9
4FWRB	WaFERS-Repair, Bypass, and Line-Patch		9
4FWLH	WaFERS-Lay Flat Hose		4
4FWTP	WaFERS-Tank Patch		0.1

UTC	Title	PAX	Short Tons
4FWFP	WaFERS-Water and Fuel Pump		6.3
4FWFF	WaFERS-Fuel Filter		5
4FWAE	RADR AC Batch Plant		35.5
4FWCR	RADR Crater Repair Equipment		302.6
4FWFC	AM-2 Crater FOD Cover		43.4
4FWCM	RADR Concrete Repair Materials		207
4FWAB	RADR Asphalt Repair Materials		230
4FWWH	RADR Warehouse Vehicles		95.6

*Under development.
Note: The PAX and short tons listed above are a template. These may vary due to individual unit LOGFORs. Units should verify PAX and weight requirements in DCAPES or the WMP-1 https://usaf.dps.mil/sites/13072/SitePages/Publications.aspx

Chapter 5

OPERATE THE AIRBASE CAPABILITIES (SUSTAIN AND RECOVER)

5.1. Overview. The Operate the Airbase (OAB) FE is designed to be IOC within seven days of arrival for a minimum duration of 30 days. The combination of the Establish the Airbase (EAB) and OAB FEs can support several Mission Generation FEs indefinitely. The OAB FE is intended to enhance combat service support capabilities. Over time, repeated loading from aircraft and/or support vehicles may cause deterioration to airfield pavement resulting in deep spalling requiring immediate repair. Spalls may also be a direct result of inadequate joint maintenance, improper construction methods, alkali- silica reactions, or munition damage. Spalls generate FOD that can cause damage to aircraft tires. In addition, damage resulting from adversarial attacks requires rapid repair to ensure airfield resiliency and mission continuation.

5.2. Sustaining Airfield Pavement. Any day-to-day airfield sustainment requirement will be determined by the structural capacity of the airfield, the deterioration rate of the airfield, condition of airfield infrastructure, volume of operations, number and quality of existing temporary repairs, and current threat data. While many airfield sustainment activities are not specifically addressed here, planners need to ensure the Senior Airfield Authority (SAA) at each operating location has sufficient forces and materials to sustain airfield operations. The Sustainment Pavement Repair (SuPR) Kit was developed to provide assets and materials to perform quality, rapid sustainment repairs that create minimal disruption to the commander's air tasking order. See AFTTP 3-32.16 for more information on sustaining airfield pavement at contingency locations. ADR capabilities to sustain airfield pavement are listed in **Table 5.1**.

Table 5.1. ADR Sustain the Airbase Capabilities.

UTC	Title	PAX	Short Tons
4F9AD	Airfield Pavement Evaluation Team	4	27.7
4F9DA	Damage Assessment		0.2
4FWSP	Sustainment, Pavement Repair Kit		58.3
****4FWPS**	ADR Paint Striping Kit		*
****4FWAE**	RADR AC Batch Plant		35.5

*Under development.
**Capabilities may be required if services are not readily available on local economy.
Note: The PAX and short tons listed above are a template. These may vary due to individual unit LOGFORs. Units should verify PAX and weight requirements in DCAPES or the WMP-1 https://usaf.dps.mil/sites/13072/SitePages/Publications.aspx

5.3. Rapid Airfield Damage Recovery. Following an airfield attack resulting in damage, the resumption of flying operations is a top priority. Engineers must be prepared to locate and assess airfield damage and explosive hazards that threaten flying operations, mitigate explosive hazards, and repair airfield pavement and infrastructure sufficiently to resume flying operations as quickly as possible. Although repair times may vary depending on force structure, available materials/ equipment, and operating environment, engineers must be able to perform standardized repairs

that support mixed rotary-wing, fighter, bomber, tanker, and airlift missions of joint aircraft directed by the Joint Force Commander. The repairs must be durable and sustainable to minimize airfield downtime. High operational tempos dictate more durable and sustainable repairs to minimize airfield downtime. RADR capabilities are listed in **Table 5.2**. **Note:** Operating in a chemical, biological, radiological, and nuclear (CBRN) environment in Mission-Oriented Protective Posture gear will extend airfield recovery timelines.

Table 5.2. RADR Recover the Airfield Capabilities.

UTC	Title	PAX	Short Tons	Source Document(s)
4FPXD	EOD Core Response Team	3*		AFTTP 3-32.5V6
4F9X1	EOD Core Equipment		2.3	CE Sup WMP-1
4FWFD	RADR FOD Removal Vehicles		123.5	
4FWCR	RADR Crater Repair Equipment		302.6	
4FWSR	Expedient Spall Repair Kit		11.5	
4FWWH	RADR Warehouse Vehicles		95.6	AFTTP 3-32.18
4FWAE	RADR AC Batch Plant		35.5	AFTTP 3-32.19
4FWCM	RADR Concrete Repair Materials		207	
4FWAB	RADR Asphalt Repair Materials		230	
4FWFC	AM2 Crater FOD Cover		43.4	
4FWFR	Fiber Reinforced Polymer Matting		24	
XFWYC	Emergency Airfield Lighting System (EALS)		15.2	T.O. 35F5-3-17-1, AFTTP 3-32.14
4FWMS	MAOS Marking Kit		*	T.O. 35E2-6-1, AFTTP 3-32.13
XFWR4	Mobile Aircraft Arresting System		29.3	T.O. 35E8-2-10-1, AFTTP 3-32.15
4FWPS	ADR Paint Striping Kit		*	T.O. 35E2-6-1, AFTTP 3-32.13
UFMAC	20-ton Tractor 6x4		30	
4FWCT	WaFERS Common Tools and Equipment		*	
4FWRB	WaFERS Repair, Bypass, and Line-Patch		*	
4FWLH	WaFERS Lay Flat Hose		*	
4FWTP	WaFERS Tank Patch		*	
4FWFP	WaFERS Water and Fuel Pump		*	
4FWFF	WaFERS Fuel Filter		*	
4FPXG	EOD Flight Leadership Element	4*		CE Sup WMP-1
4FPXC	EOD Leadership Support Element	4*		CE Sup WMP-1
4F9X3	EOD Base Support Sustainment Equip.		3.1	CE Sup WMP-1

4F9X5	EOD Explosive Storage Kit		9.4	CE Sup WMP-1
4F9X8	EOD Demolition Materials		5.8*	CE Sup WMP-1
4FWE1	EOD Vehicle, MRAP, Cougar Cat I		17.9	CE Sup WMP-1

*Under development.
Note: The PAX and short tons listed above are a template. These may vary due to individual unit LOGFORs. Units should verify PAX and weight requirements in DCAPES or the WMP-1 https://usaf.dps.mil/sites/13072/SitePages/Publications.aspx

Chapter 6

ROBUST THE AIRBASE CAPABILITIES

6.1. Overview. Robust the Airbase FE allows support forces to enhance capabilities already in place from previous FEs until a rotational operation can be implemented. These forces typically do not arrive until 30 days after an operating location is established.

6.2. Capabilities. Specific ADR Capabilities within this FE expand the MAOS, provide expeditionary trim pads and aircraft tie-downs, upgrade expeditionary/temporary arresting systems, and upgrade crushed stone repairs with semi-permanent caps to help prevent FOD hazards. These capabilities are identified in **Table 6.1**.

Table 6.1. ADR Capabilities to Robust the Airfield.

UTC	Title	PAX	Short Tons
4F9DA	Damage Assessment		0.2
4FWAM	WRM Expansion AM-2 Mat & Support Equipment		154.5
N/A	**Expedient Trim Pad Kit**		
N/A	**Expedient Aircraft Tie-Down Anchors**		
4FWAE	RADR AC Batch Plant		35.5
4FWCR	RADR Crater Repair Equipment		22.4
4FWCM	RADR Concrete Repair Materials		207
4FWAB	RADR Asphalt Repair Material		230
4FPXG	EOD Flight Leadership Element	4	
4FPXC	EOD Leadership Support Element	4	
4F9X3	EOD Base Support Sustainment Equipment		3.1
4F9X5	EOD Explosive Storage Kit		9.4
4F9X8	EOD Demolition Materials		5.8*
4FWE1	EOD Vehicle, MRAP, Cougar Cat I		17.9
Note: The PAX and short tons listed above are a template. These may vary due to individual unit LOGFORs. Units should verify PAX and weight requirements in DCAPES or the WMP-1 https://usaf.dps.mil/sites/13072/SitePages/Publications.aspx			

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 10-202, *Contingency Response Forces*, 11 June 2020

AFI 10-209, *RED HORSE Program*, 11 June 2019

AFI 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*, 25 October 2023

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

CE Supplement to the War and Mobilization Plan-1 (WMP-1), 13 April 2020

T.O. 35E2-2-7-WA-1, *AM-2 Airfield Landing Mat and Accessories*, 6 September 1985

T.O. 35E2-5-1-WA-1, *Crushed Stone Crater Repair and Line-of-Sight Profile Measurement for Rapid Runway Repair*, 27 August 2007

T.O. 35E2-6-1, *Minimum Airfield Operating Surface Marking System (MAOSMS)*, 4 November 2013

T.O. 35E8-2-10-1, *Arresting Systems, Aircraft, Mobile*, 24 July 2022

T.O. 35F5-3-17-1, *Lighting System, Airfield, Emergency A/E82U-2*, 12 May 2019

TSPWG M 3-270-01.3-270-07, *Airfield Damage Repair*, 21 May 2020

UFC 3-250-03, *Standard Practice Manual for Flexible Pavements,* 30 May 2018

UFC 3-250-08FA, *Standard Practice for Sealing Joints and Cracks in Rigid and Flexible Pavements*, 16 January 2004

UFC 3-250-09FA, *Aggregate Surfaced Roads and Airfields Areas*, 16 January 2004

UFC 3-250-11, *Soil Stabilization and Modification for Pavements*, 30 November 2020

UFC 3-260-01, *Airfield and Heliport Planning and Design*, 4 February 2019

UFC 3-260-02, *Pavement Design for Airfields*, 30 June 2001

UFC 3-260-03, *Standard Practice for Airfield Pavement Evaluation*, 21 August 2023

UFC 3-260-04, *Airfield and Heliport Marking*, 16 May 2018

UFC 3-260-11FA, *Model Design-Build (D-B) Request for Proposal (RFP) For Airfield Contracts*, 25 May 2005

UFC 3-260-16, *O&M Manual: Standard Practice for Airfield Pavement Condition Surveys*, 3 February 2019

UFC 3-260-17, *Dust Control for Roads, Airfields and Adjacent Areas*, 22 May 2018

UFC 3-270-01, *O&M Manual: Asphalt and Concrete Pavement Maintenance and Repair*, 21 February 2018

AFTTP 3-32.5, Volume 6, *Explosive Ordnance Disposal (EOD) Unexploded Explosive Ordnance (UXO) Response Operations*, 11 October 2023

AFTTP 3-32.10, *Introduction to Rapid Airfield Damage Recovery (RADR),* 15 October 2019

AFTTP 3-32.11, *Airfield Damage Assessment after Major Attack*, 18 May 2022

AFTTP 3-32.12, *Minimum Airfield Operating Surface (MAOS) Selection and Repair Quality Criteria (RQC)*, 3 May 2024

AFTTP 3-32.13, *Airfield Marking and Striping after Major Attack*, 8 December 2023

AFTTP 3-32.14, *Alternate Installation Sequence for Emergency Airfield Lighting System after Major Attack*, 19 January 2024

AFTTP 3-32.15, *Rapid Setback Installations of Mobile Arresting Gear System*, 21 June 2016

AFTTP 3-32.16, *Sustaining Airfield Pavement at Enduring Contingency Locations*, 20 January 2017

AFTTP 3-32.17, *Rapid Airfield Crater and Spall Repair*, 10 October 2023

AFTTP 3-32.18, *Rapid Airfield Damage Repair Warehouse Operations*, 9 February 2024

AFTTP 3-32.19, *Rapid Airfield Damage Repair Asphalt Batch Plant Operations*, 24 September 2018

AFTTP 3-32.22, *Water and Fuel Expedient Repair System (WaFERS)*, 12 February 2024

Joint Publication 3-30, *Joint Air Operations,* 25 July 2019

AFDP 3-34, *Engineer Operations*, 6 October 2021

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*, 15 April 2022

Abbreviations and Acronyms

ADR—Airfield Damage Repair

AFCEC—Air Force Civil Engineer Center

AFI—Air Force Instruction

AFFORGEN—Air Force Force Generation

AFTTP—Air Force Tactics, Techniques, and Procedures

AM-2—Aluminum Matting, Version 2

BEEF—Base Engineer Emergency Force

C2—Command and Control

CBRN—Chemical, Biological, Radioactive, and Nuclear

CE—Civil Engineer

CRE—Contingency Response Element

DCAPES—Deliberate and Crisis Actions Planning Segments

DFT—Demand Force Teams

EAB—Establish the Airbase

EALS—Emergency Airfield Lighting System

EOD—Explosive Ordnance Disposal

FE—Force Element

FOD—Foreign Object Debris

FRP—Fiber Reinforced Polymer

IOC—Initial Operating Capability

MAOS—Minimum Airfield Operating Surface

MOS—Minimum Operating Strip

OAB—Operate the Airbase

OPR—Office of Primary Responsibility

PAX—Passengers

RADR—Rapid Airfield Damage Recovery

SAA—Senior Airfield Authority

SuPR—Sustainment, Pavement Repair

T.O.—Technical Order

TTP—Tactics, Techniques, and Procedures

UTC—Unit Type Code

UXO—Unexploded Explosive Ordnance

WaFERS—Water and Fuel Expedient Repair System

WMP-1—War and Mobilization Plan, Volume-1

WRM—War Reserve Materiel

Office Symbols

AF/A4CX—Air Force Directorate of Civil Engineers, Readiness Division

AFCEC—Air Force Civil Engineer Center

Terms

Airfield—An area prepared for the accommodation (including any buildings, installations, and equipment), landing, and takeoff of aircraft.

Air Force Force Generation (AFFORGEN)—AFFORGEN is the Air Force's sustainable, capacity-driven model for presenting forces to Joint Force Commanders. AFFORGEN's intent is to enable operational preparedness and readiness recovery to compete with peer competitors, while clearly focusing USAF efforts on a predictable and sustainable force offering. All MAJCOMs will align their forces into the AFFORGEN model.

Air Tasking Order—A method used to task and disseminate to components, subordinate units, and command and control agencies projected sorties, capabilities and/or forces to targets and specific missions. Normally provides specific instructions to include call signs, targets, controlling agencies, etc., as well as general instructions. Source: Joint Publication 3-30, *Joint Air Operations*.

Command and Control (C2)—The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of the mission. C2 functions are performed through an arrangement of personnel, equipment, communications, facilities, and procedures employed by a commander in planning, directing, coordinating, and controlling forces and operations in the accomplishment of the mission.

Contingency—An emergency involving military forces caused by natural disasters, terrorists, subversives, or by required military operations. Due to the uncertainty of the situation, contingencies require plans, rapid response and special procedures to ensure the safety and readiness of personnel, installations, and equipment.

Crater—The pit, depression, or cavity formed in the surface of the earth by an explosion. It may range from saucer-shaped to conical, depending largely on the depth of burst.

Damage Assessment—1. The determination of the effects that attacks have on targets. 2. (DOD only) A determination of the effect of a compromise of classified information on national security. 3. (AF/CE) The process of identifying and locating damage and unexploded ordnance following an attack. Damage assessment activities generally are separated into two categories: airfield pavements and facility/utility.

Explosive Hazard—Any hazard containing an explosive component to include unexploded explosive ordnance (including land mines), booby traps (some booby traps are nonexplosive), improvised explosive devices (which are an improvised type of booby trap), captured enemy ammunition, and bulk explosives.

Explosive Ordnance—All munitions containing explosives, nuclear fission or fusion materials, and biological and chemical agents.

Explosive Ordnance Disposal (EOD)—The detection, identification, on-site evaluation, rendering safe, recovery, and final disposal of unexploded explosive ordnance. It may also include explosive ordnance which has become hazardous by damage or deterioration.

Facility—A real property entity consisting of one or more of the following: a building, a structure, a utility system, pavement, and underlying land.

Minimum Operating Strip (MOS)—1. A runway which meets the minimum requirements for operating assigned and/or allocated aircraft types on a particular airfield at maximum or combat gross weight. 2. The MOS is the smallest amount of area to be repaired to launch and recover aircraft after an attack. Selection of this MOS will depend upon mission requirements, taxi access, resources available, and estimated time to repair. For fighter aircraft, the typically accepted dimensions are 5,000 feet long by 50 feet wide.

Mitigation—Activities designed to reduce or eliminate risks to persons or property or to lessen the actual or potential effects or consequences of an incident. Mitigation measures may be implemented prior to, during, or after an incident. Mitigation measures are often developed in accordance with lessons learned from prior incidents. Mitigation involves ongoing actions to reduce exposure to, probability of, or potential loss from hazards. Measures may include zoning

and building codes, flood plain buyouts, and analysis of hazard-related data to determine where it is safe to build or locate temporary facilities. Mitigation can include efforts to educate governments, businesses, and the public on measures they can take to reduce loss and injury.

Ordnance—Explosives, chemicals, pyrotechnics, and similar stores, e.g., bombs, guns and ammunition, flares, smoke, or napalm.

Personnel—Those individuals required in either a military or civilian capacity to accomplish the assigned mission.

Recovery—The development, coordination, and execution of service- and site-restoration plans for impacted communities and the reconstitution of government operations and services through individual, private-sector, nongovernmental, and public assistance programs that: identify needs and define resources; provide housing and promote restoration; address long-term care and treatment of affected persons; implement additional measures for community restoration; incorporate mitigation measures and techniques, as feasible; evaluate the incident to identify lessons learned; and develop initiatives to mitigate the effects of future incidents.

RED HORSE—Air Force units wartime-structured to provide a heavy engineer capability that are mobile, rapidly deployable, and largely self-sufficient for limited periods of time.

Runway—A defined rectangular area of an airfield, prepared for the landing and takeoff run of aircraft along its length. A runway is measured from the outer edge of the thresholds from one end of the runway to the others. The width of the runway is typically measured from the outer edge of the load-bearing pavement on one side to the outer edge of the load-bearing pavement on the other side. In some cases the runway may be measured from the outside edge of the runway marking line on one side to the outside edge of the marking line on the other side and any remaining load bearing pavement is considered shoulder.

Senior Airfield Authority (SAA)—An individual designated/appointed by the component responsible for airfield operations at direction of the Joint Force Commander. The SAA is responsible for control, priorities, operation and maintenance of an airfield to include runways, associated taxiways, parking ramps, land, and facilities whose proximity affect airfield operations.

Spall—Pavement damage that does not penetrate through the pavement surface to the underlying base course layers. A spall damage area could be up to 1.5 meters (5 feet) in diameter.

Taxiway—A specially-prepared or designated path on an airfield or heliport, other than apron areas, on which aircraft move under their own power to and from landing, takeoff, service, and parking areas.

Techniques—Non-prescriptive ways or methods use to perform missions, functions, or tasks.

Threat—An indication of possible violence, harm, or danger.

Unexploded Explosive Ordnance (UXO)—Explosive ordnance which has been primed, fused, armed, or otherwise prepared for action, and which has been fired, dropped, launched, projected, or placed in such a manner as to constitute a hazard to operations, installations, personnel, or material, and remains unexploded either by malfunction or design.

Attachment 2

ENGINEER REACHBACK AND OTHER USEFUL LINKS

Table A2.1. Engineer References and Resources Links.

Engineer Reachback
Air Force Civil Engineer Center (AFCEC): https://www.afcec.af.mil/
CE DASH (AFCEC Technical Support Portal): https://usaf.dps.mil/teams/CEDASH/scripts/homepage/home.aspx
CE Playbooks: https://www.ceplaybooks.com
AF Publications and Forms: https://www.e-publishing.af.mil/
AFCEC Reachback Center: email at AFCEC.RBC@us.af.mil
My Learning (Learning Management System): https://lms-jets.cce.af.mil/moodle/login/index.php
AF Design Guides (AFDG): https://www.wbdg.org/ffc/af-afcec
DOD Supplemental Technical Documents (TSPWG M): https://www.wbdg.org/ffc/dod/supplemental-technical-documents
Whole Building Design Guide (WBDG): https://www.wbdg.org/
US Army Corp of Engineers Official Publications, http://www.publications.usace.army.mil/Home.aspx
Unified Facilities Criteria (UFC): https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc
Unified Facilities Guide Specifications (UFGS): https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs
USACE Reachback Operations Center (UROC): https://uroc.usace.army.mil
USACE Protective Design Center: https://intelshare.intelink.gov/sites/pdc/SitePages/Home.aspx
Army Publications and Forms: https://armypubs.army.mil/
Navy Doctrine Library System: https://doctrine.navy.mil/default.aspx
DOD Issuances: https://www.esd.whs.mil/DD/DoD-Issuances/
Joint Publications: https://jdeis.js.mil/my.policy
Armed Forces Pest Management Board: https://www.acq.osd.mil/eie/afpmb/